

Status Fitur ACS

Daftar fitur · status backend · ketersediaan UI · integrasi & pengujian

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Legenda kolom:

Backend / UI / Integrasi & Uji: ✓ lengkap • sebagian ✗ belum — tidak berlaku

Integrasi & Uji: "e2e S#" = tercakup skenario end-to-end otomatis (10/10 lulus, lihat PENGUJIAN_E2E.md). "lapangan" = menunggu uji CPE fisik. "browser" = menunggu uji UI di browser.

CATATAN PENTING

Hampir semua UI **belum diverifikasi visual di browser sungguhan** — uji `ui-qa-browser` belum tuntas. "✓" pada kolom UI berarti halaman/komponen & wiring API-nya *ada dan lulus build*, bukan "sudah diklik & dikonfirmasi tampil benar".

1. Autentikasi & Keamanan

Fitur	Backend	UI	Integrasi & Uji	Catatan
Login username/password (JWT)	✓	✓	e2e (semua skenario)	JWT 8 jam; LoginPage
Proteksi brute-force (5× → kunci 15 mnt)	✓	—	unit test	pesan error disamarkan
Rate limit berlapis (login 2/s, CWMP 5/s, REST 30/s)	✓	—	e2e	in-memory per-IP — ganti Redis sebelum multi-instance
Ganti password sendiri	✓	✓	e2e S10	ProfilePage · PATCH /auth/password
Admin reset password user lain	✓	✓	•	Administration → Users
SSO / OIDC	•	•	✗	state / nonce belum divalidasi (LAPORAN §4.3)
API Token M2M (terbit / daftar / cabut)	✓	✓	e2e	Administration → API Token; Bearer <code>acs_...</code>
Batas ukuran body (anti-DoS)	✓	—	e2e (regresi)	REST 2 MiB, CWMP 4 MiB
SSRF guard (Connection Request URL & webhook)	✓	—	e2e S10 + unit	<code>pkg/netguard</code> + <code>host==RemotelP</code>

Enkripsi kredensial at-rest (AES-256-GCM)	✓	—	✓	shared secret CWMP, connection-request password
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2. Multi-tenant & IAM

Fitur	Backend	UI	Integrasi & Uji	Catatan
Tenant CRUD + aktif/nonaktif	✓	✓	e2e S6/S10	Administration → Tenants; nonaktif = token langsung ditolak
Shared secret Inform CWMP per tenant	✓	✓	e2e S1	Administration → Kredensial CWMP
Wizard onboarding tenant	✓	✓	• browser	/administration/onboarding, 3 langkah
User CRUD + kelola peran (RBAC 5 role)	✓	✓	e2e S6/S10	SUPERADMIN / ADMIN / NOC / VIEWER / ENDUSER
Isolasi data antar-tenant	✓	—	e2e S6	admin tenant B tidak lihat device tenant A
ENDUSER dikurung ke /self-service	✓	✓	e2e S10	middleware staffOnly — 403 di semua API staf
White-labeling (logo / nama / warna aksen)	✓	✓	• browser	Administration → Branding; ADMIN self-service
Kuota task queue per tenant	✓	✓	e2e (kuota)	Administration → Kuota Task; superadmin only

3. Inventory & Monitoring Perangkat

Fitur	Backend	UI	Integrasi & Uji	Catatan
Onboarding device dari Inform	✓	✓	e2e S1 (5 vendor)	tenant & vendor auto-resolve
Daftar device + filter & pencarian + seleksi massal	✓	✓	e2e	DevicesPage ; filter vendor/status/tag/search
Detail device + tree parameter TR-069	✓	✓	•	expand / edit / AddObject / DeleteObject (ParameterTree)
Histori event CWMP	✓	✓	e2e S1/S7	tab "Histori Event"
Nilai parameter tersimpan (EAV)	✓	✓	e2e S4	tab "Parameter"
Capture ConnectionRequestURL otomatis	✓	—	e2e S7	dari parameter Inform; anti-SSRF (host==RemoteIP)

Config snapshot device (ambil / riwayat)	✓	✓	●	tab "Config History"
Geolokasi device (lat / long)	✓	●	●	field ada (PATCH device); tampilan peta belum
Redaman optik RX/TX power (ONT GPON)	●	✓	X lapangan	tab "Redaman Optik"; mapping optical per vendor kosong
Status online/offline + reaper sesi basi	✓	✓	✓	sweeper 15 menit
Deteksi anomali (DNS hijack, reboot loop, redaman kritis)	✓	●	X lapangan	tercatat di <code>activity_logs</code> + webhook

4. Protokol CWMP / TR-069

Fitur	Backend	UI	Integrasi & Uji	Catatan
Inform semua event code (0/1/2/4/6/7)	✓	—	e2e S1/S7	string spec Broadband Forum
Auth Inform (shared secret + anti-spoof lintas tenant)	✓	—	e2e S1	+ orphan healing device lama
Echo namespace CWMP (cwmp-1-0 / 1-1 / 1-2)	✓	—	e2e S1	5 profil vendor beda namespace
Root data model TR-098 & TR-181	✓	✓	e2e S1	resolve per device, tidak diasumsikan
RPC Get / SetParameterValues	✓	✓	e2e S2/S4	+ chunking proaktif 50 nama/task
RPC GetParameterNames + auto-discovery	✓	✓	●	refresh tree; auto utk vendor tak dikenal saat BOOTSTRAP
RPC AddObject / DeleteObject	✓	✓	●	tombol di <code>ParameterTree</code>
RPC Reboot	✓	✓	e2e S3	Device Detail + bulk
RPC FactoryReset	✓	✓	●	Device Detail (ADMIN)
RPC Download (firmware) + TransferComplete	✓	✓	e2e S8	presigned URL MinIO
Fault handling: 9005 gagal permanen, 9003 split	✓	—	e2e S5	tidak retry percuma
Connection Request HTTP (ACS → CPE)	✓	✓	e2e S7	502 (bukan 500) bila CPE offline
Connection Request UDP / STUN (TR-111, CGNAT)	●	—	X lapangan	kode ada, belum teruji ke CGNAT nyata

Sesi stateful multi-instance (state di Redis + MariaDB)	✓	—	e2e	app server horizontal-scale
Logging terstruktur jalur CWMP (slog JSON)	✓	—	✓	+ metrik latensi Inform→InformResponse

5. Task / Antrean RPC

Fitur	Backend	UI	Integrasi & Uji	Catatan
Antrean task per device + prioritas	✓	✓	e2e	TasksPage ; index (device_id,status,priority)
Retry + max_retries + backoff eksponensial	✓	✓	unit + e2e S5	
Batalkan task	✓	✓	•	TasksPage
Statistik task (dashboard)	✓	✓	✓	agregat tenant-scoped
Redaksi URL/kredensial di task utk role < ADMIN	✓	✓	✓	NOC/VIEWER dapat [redacted]

6. Provisioning (Profile + Zero-Touch)

Fitur	Backend	UI	Integrasi & Uji	Catatan
Provisioning Profile CRUD	✓	✓	e2e S2	ProvisioningPage
Parameter picker (autocomplete logical key)	✓	✓	• browser	preview tr069_path hasil resolve
Apply profile (1 device / massal)	✓	✓	e2e S2	Device Detail + Devices bulk
Zero-Touch Rule CRUD	✓	✓	•	ZTRuleModal di ProvisioningPage
ZTP match: OUI / serial / vendor / model / versi FW / nilai param	✓	✓	•	SQL LIKE wildcard sudah diperbaiki
ZTP aksi gabungan (profile + reboot + firmware)	✓	✓	•	
ZTP trigger BOOTSTRAP_ONLY / BOOT / EVERY_INFORM	✓	✓	•	+ pagar anti-reboot-loop
ZTP gagal → tercatat (bukan gagal senyap, FR-15)	✓	•	✓	ZERO_TOUCH_APPLY_FAILED di activity_logs

7. Preset Engine (penegakan konfigurasi)

Fitur	Backend	UI	Integrasi & Uji	Catatan
Preset CRUD (precondition + configurations JSON)	✓	✓	e2e S9	PresetsPage ; editor JSON
Enforce=1 drift-heal tiap Inform	✓	✓	e2e S9	device menyimpang → auto SetParameterValues
Pagar: cooldown 15m / maks 3 gagal / tunggu task in-flight	✓	—	unit	
Jitter PeriodicInformInterval (anti thundering-herd)	✓	—	✓	offset deterministik per device
Estimasi "N device terpengaruh" saat aktifkan enforce	✗	✗	—	v1.1 (butuh endpoint count)
Op apply_profile / refresh , precondition tag_id / channel	✗	✗	—	v1.1; v1 hanya set_parameter

8. Firmware

Fitur	Backend	UI	Integrasi & Uji	Catatan
Upload file firmware (multipart, checksum SHA-256 server-side)	✓	✓	e2e S8	superadmin only
Object storage MinIO + presigned URL ke CPE	✓	—	e2e S8	URL 1 jam, digenerate on-demand
Upgrade 1 device	✓	✓	•	Device Detail → Firmware
Canary / staged rollout batch (wave % + failure threshold)	✓	✓	e2e S8	Firmware → Rollout Batches
Auto-advance wave / advance manual / cancel / auto-pause	✓	✓	e2e S8	PAUSED bila failure rate > ambang
Firmware jobs per device	✓	✓	e2e S8	
Sweeper objek yatim MinIO	✗	—	—	follow-up (LAPORAN)

9. Diagnostics

Fitur	Backend	UI	Integrasi & Uji	Catatan
Trigger diagnostic (ping / traceroute / dll)	✓	✓	• lapangan	tab "Diagnostics"

Hasil diagnostic	✓	✓	● lapangan	korelasi ke task GetParameterValues
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10. Katalog Vendor

Fitur	Backend	UI	Integrasi & Uji	Catatan
Vendor CRUD	✓	✓	e2e (add OUI)	5 vendor: ZTE/Huawei/FiberHome/Nokia/Cdata
OUI per vendor	✓	✓	e2e S1	data kosong — isi dari IEEE registry sebelum uji lapangan
Device Model	✓	✓	●	0 baris (butuh sample device nyata)
Vendor Parameter Mapping (logical key → path TR-069)	✓	✓	●	120 baris seeded (migrasi 0004): 12 key × 5 vendor × TR-098/181
Playbook onboarding vendor baru	✓	—	—	VENDOR_ONBOARDING.md

11. Tags & Segmentasi

Fitur	Backend	UI	Integrasi & Uji	Catatan
Tag CRUD	✓	✓	e2e S10 (create)	TagsPage
Assign / remove tag ke device	✓	✓	●	Device Detail
Filter device by tag	✓	✓	●	GET /devices?tag_id= — butuh index (tag_id,device_id) di skala besar

12. Audit Trail & Observability

Fitur	Backend	UI	Integrasi & Uji	Catatan
Audit trail 7-kolom (tabel master)	✓	—	✓	created/updated/deleted by+at
Feed activity log global ("siapa mengubah apa")	✓	✓	●	AuditPage (ADMIN+), tenant-scoped
Timeline audit per device	✓	✓	✓	tab "Timeline Audit"

Metrik Prometheus (6 metrik + histogram latensi)	✓	●	✓	/metrics tanpa auth (firewall di jaringan)
Alert rules (backlog PENDING, task FAILED per vendor)	✓	●	●	prometheus_alerts.yml ; Alertmanager belum kirim keluar
Dashboard Grafana visual	✗	✗	—	hanya UI bawaan Prometheus
Runbook operasional insiden	✓	—	—	RUNBOOK.md

13. Integrasi Sistem Lain

Fitur	Backend	UI	Integrasi & Uji	Catatan
REST API internal (103 endpoint, RBAC + tenant-scope)	✓	✓	e2e	parity backend↔frontend 95/95 relevan
Spesifikasi OpenAPI 3.0.3	✓	—	redocly lint	openapi.yaml 1:1 dengan router.go
Koleksi Postman (auto-generate)	✓	—	Newman	ACS-API.postman_collection.json
API token untuk integrasi machine-to-machine	✓	✓	e2e	ter-scope tenant + peran, bisa dicabut
Webhook keluar (typed event, HMAC-SHA256)	✓	✓	●	create/list/test/log delivery UI; SSRF guard
WebSocket push event (realtime dashboard)	●	●	✗	hub in-memory — parsial di multi-instance (LAPORAN C5)
Prometheus scrape endpoint	✓	—	✓	
USP / TR-369	✗	✗	—	hanya kerangka (HandleMessage no-op) — tidak operasional
Notifikasi email / WhatsApp	✗	✗	—	tidak ada di lingkup proyek ini

14. Portal Self-Service (ENDUSER)

Fitur	Backend	UI	Integrasi & Uji	Catatan
List perangkat milik pelanggan	✓	✓	e2e S10	via user_devices

Ubah SSID / passphrase / band WiFi	✓	✓	● browser	SelfServicePage
Restart perangkat sendiri	✓	✓	● browser	
Isolasi: hanya perangkat sendiri (bukan se-tenant)	✓	—	e2e S10	

15. UI/UX Umum

Fitur	Backend	UI	Integrasi & Uji	Catatan
Dashboard analitik (Recharts: status, vendor, task, redaman optik)	✓	✓	● browser	DashboardPage
Dark mode (3-state: sistem / terang / gelap)	—	✓	✗ browser	belum diverifikasi visual
Tampilan mobile-responsive (sidebar drawer)	—	✓	✗ browser	belum diverifikasi visual
Command palette (Ctrl/Cmd+K)	—	✓	● browser	cari device + navigasi
Notifikasi in-app (bell + panel)	—	✓	● browser	polling (bukan push realtime)
i18n (Bahasa Indonesia / Inggris)	—	✓	● browser	cakupan awal, fallback ke ID
Bulk actions (apply profile / firmware / reboot ke banyak device)	✓	✓	●	device vendor tak cocok dilewati otomatis
Halaman Profil user	✓	✓	●	ProfilePage

16. Ringkasan

Kategori	Backend lengkap	Ada UI	Tercakup e2e
Autentikasi & Keamanan	8 / 10	6 / 10	7 / 10
Multi-tenant & IAM	8 / 8	8 / 8	7 / 8
Inventory & Monitoring	9 / 11	10 / 11	6 / 11
Protokol CWMP	13 / 15	9 / 9 relevan	10 / 15
Task queue	5 / 5	5 / 5	4 / 5
Provisioning	8 / 8	8 / 8	3 / 8

Preset engine	4 / 6	4 / 6	2 / 6
Firmware	6 / 7	6 / 7	5 / 7
Katalog vendor	5 / 5	4 / 4 relevan	2 / 5
Audit & Observability	6 / 7	5 / 7	3 / 7
Integrasi eksternal	6 / 9	5 / 9	5 / 9
Portal self-service	4 / 4	4 / 4	2 / 4

Yang benar-benar belum ada (backend + UI): USP/TR-369 operasional, dashboard Grafana, estimasi "N device" preset, op preset `apply_profile`, sweeper objek yatim firmware, Alertmanager notifikasi keluar, notifikasi email/WA.

Ada tapi butuh pekerjaan: OIDC (validasi state), WebSocket multi-instance (Redis pub/sub), rate limiter terdistribusi, redaman optik (mapping vendor), CGNAT/STUN (uji lapangan).

Verifikasi yang tertunda: uji UI di browser sungguhan (semua halaman), uji CPE fisik (interoperabilitas firmware vendor), load test kapasitas.